Assignment Handout

Competitive Data-Center Connection Planning Game

Course name

October 8, 2026

Assignment in one sentence. You operate a data-center, choose where and how much load to connect in a power grid, optionally build private generation and storage, and maximize your profit over a six-month hourly operating horizon.

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1 Assignment overview

Each student group represents a data-center operator such as a cloud-computing company. Your group submits a connection-and-resource plan. Submissions of all groups will be collected and evaluated together using a DC optimal power flow (`DCOPF`) dispatch model. **Groups are ranked by score.**

1.1 What you must do

Your team must design and submit one complete half-year connection plan in the released CSV format. The goal is to serve valuable computational load while controlling energy, interconnection, generation, and storage costs.

1.2 Deliverables

1. one `UTF-8 CSV` file for your team;
2. validator output showing that the canonical file passes the checker;
3. a short strategy note explaining your main bus, load, generator, and storage choices and the trade-offs you considered;
4. the exact final CSV and any code used to generate it.

The instructor will announce the file name, submission location, deadline, round assignment, and whether the strategy note is required for grading. Do not submit a screenshot in place of the CSV.

1.3 Assignment rules at a glance

- Submit one canonical site-level CSV row for each selected site. The CSV has no hourly period, load, or horizon rows.
- Use `bus IDs` from the supplied ACTIVSg2000 scenario block. Site capacities are specified in MW/MWh.
- The released calendar still determines the evaluator's half-year hourly scenario: 2028 has 4,368/4,416 hours and 2027/2029 have 4,344/4,416 hours for January–June/July–December. Those hours are not CSV rows.
- `Generators` and `storage` are private to the submitting team.
- The current release has no student-controlled `curtailment` decision.
- The instructor evaluates valid teams jointly and ranks teams by `score`.

Use of AI. You are explicitly encouraged to use available AI tools while doing the assignment, including ChatGPT, Codex, Claude Code, GitHub Copilot, and similar tools. You remain responsible for checking generated code, understanding your submission, respecting the CSV contract, and explaining your engineering decisions. AI assistance is part of the intended computational workflow, not a substitute for validation.

1.4 Learning objectives

- translate a planning idea into a machine-readable decision file;
- reason about spatial congestion, generator dispatch, and storage;
- understand why individually feasible plans can be jointly infeasible;
- use reproducible computational experiments and inspect diagnostics;
- communicate trade-offs among IT value, energy cost, and investment.

2 Data and model: what is real and what is instructional

The dispatch input is the public synthetic Texas ACTIVSg2000 model [1], which contains exactly **2,000 buses**, 3,206 branches, and 544 generator records. The fixed topology is combined with a controlled three-year hourly scenario package. The scenario provides deterministic bus-level load; active generator status, PMIN, and PMAX values come directly from the dataset. Because ACTIVSg2000 does not provide technology labels or hourly availability data, the active release does not apply synthetic wind/solar deratings. Each block is validated hour by hour with DCOPF. It is an instructional scenario, not historical ERCOT telemetry or an official ERCOT operating model.

The conceptual pipeline is:

Public ACTIVSg2000 topology + controlled hourly operating profiles
⇓
Fixed ACTIVSg2000 buses, branches, generators, costs, and scenario demand
⇓
DCOPF input and teaching scoring scenario
⇓
Joint evaluation of student CSV files

The following distinction is important:

- **Public data:** `bus`, `branch`, `generator`, and marginal-cost records from the ACTIVSg2000 source model.
- **Instructor preparation:** `formatting`, `validation`, and the full-horizon economic profiles supplied with the round.
- **Instructional assumptions:** the student-owned investments, connection requests, `DCOPF` treatment, `IT value`, capital-cost coefficients, storage-cost coefficients, and the `fair continuous projection` policy.

The exercise should not be interpreted as an ERCOT planning recommendation or as a forecast of actual interconnection feasibility.

3 Operating horizon and decisions

Each submission governs **one half-year with a one-hour time step:**

Year	Round	Hours	Zero-based period indices
2028	January–June	4,368	$0, \ldots, 4367$
2028	July–December	4,416	$0, \ldots, 4415$
2027 or 2029	January–June	4,344	$0, \ldots, 4343$
2027 or 2029	July–December	4,416	$0, \ldots, 4415$

The source year contains one full calendar year's hourly records. A submission therefore has access to roughly 2,000 possible buses and roughly 4,400 hourly evaluation states, but it does *not* encode those states as CSV rows. The canonical CSV lists only the sites used by your team; the evaluator combines those site decisions with the selected hourly scenario and economic artifacts.

For each selected site s, submit a nonnegative connection capacity and optional private generation and storage capacities. A team's resources serve only that team. They cannot export energy to another team or become a shared pool. The `bus_id` must come from the supplied bus catalog.

4 Mathematical model

4.1 Connection load

Let $C_{i,s} \geq 0$ be the connection capacity submitted by team i for site s. The canonical CSV stores $C_{i,s}$ directly, together with optional generator and storage capacities; it does not store an hourly load profile. The evaluator uses the site-level plan when constructing each hourly dispatch.

4.2 Private generation

Each positive generator capacity in the canonical CSV represents a team-built generator. The seven-column CSV has no generator- or storage-ID columns; the evaluator assigns deterministic private asset IDs internally from the team and site IDs. These IDs are not lookup keys into the public generator catalog. The evaluator applies the same convex piecewise-linear operating-cost curve to every team-built generator at every site: the first 25% of capacity costs \$12/MWh, the next 50% costs \$16/MWh, and the final 25% costs \$24/MWh. You submit the capacity and the evaluator chooses hourly dispatch. For a generator with capacity $G_{i,g}$, dispatch satisfies

$$0 \leq g_{i,g,t} \leq G_{i,g}.$$

The generator is physically attached to the submitted bus and is private to your team. It is not an existing public generator and does not inherit a public generator ID, PMIN, PMAX, or marginal cost.

4.3 Private storage

For storage device b, submit power capacity $P_{i,b}$ in MW and energy capacity $B_{i,b}$ in MWh. The evaluator assigns the device's deterministic private asset ID internally; the student CSV contains no `storage_id` field. Charging $c_{i,b,t}$ and discharging $q_{i,b,t}$ are continuous. We assume ideal charging/discharging efficiency:

$$e_{i,b,t} = e_{i,b,t-1} + c_{i,b,t} - q_{i,b,t},$$

with

$$0 \le e_{i,b,t} \le B_{i,b}, \qquad 0 \le c_{i,b,t}, q_{i,b,t} \le P_{i,b},$$
$$q_{i,b,t} \le e_{i,b,t-1}, \qquad e_{i,b,0} = 0.5B_{i,b}, \qquad e_{i,b,H} \ge 0.5B_{i,b}.$$

Storage is private to the submitting team. Charging appears as additional team demand and discharging reduces that team's grid purchase; it does not serve any other team's load.

Storage cost calibration. The released half-year scoring files use the following instructional coefficients:

$$C^{\text{storage,cap}}_{i,b} = (\$200{,}000/\,\text{MW})P_{i,b} + (\$300{,}000/\,\text{MWh})B_{i,b},$$
$$C^{\text{storage,op}}_{i,b} = (\$8/\,\text{MWh}) \times \text{discharged energy}.$$

For a four-hour device, the two capital terms correspond to approximately \$1,400/kW of installed power capacity. These are rounded teaching coefficients informed by U.S. battery-storage construction and O&M benchmarks, not a project quotation or an ERCOT tariff [9, 6, 5]. The same coefficients are used for both half-year rounds unless the instructor announces a revised round file. The reference literature also reports fixed O&M near \$40/kW-year; the current evaluator does not have a separate annual fixed-O&M field, so that value is documented as a reference and is not charged in addition to the \$8/MWh operating proxy.

4.4 DCOPF

For each hour, the instructor solves a linear `DCOPF` on the prepared Texas network. Network balance, generator limits, branch thermal limits, and fixed base demand are enforced. Load is inelastic in the ordinary feasible case: there is no integer unit commitment and no ordinary load-shedding variable.

5 CSV submission format

Submit one `UTF-8 CSV` file per team. The canonical header is exactly:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
```

Each data row is one site. The file has no `period`, hourly `load_mw`, `submission_id`, `generator_id`, or `storage_id` columns and contains no hourly horizon rows. The `site_id` is a team-defined site label; `bus_id` must come from the supplied scenario block.

Example:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
team-blue,site-north,1001,40,20,10,40
team-blue,site-south,1002,15,0,0,0
```

Values use MW, MWh, and decimal notation. The `validator` checks the exact seven-column header, site identifiers, scenario bus identifiers, finite nonnegative capacities, and the storage ratio.

Before submitting, run the provided `CSV validator`. From the project root, run with the released ACTIVSg2000 scenario-block path:

```
PYTHONPATH=src python scripts/check_submission.py \\
  --file team-blue.csv
```

The `checker` is also available after installation as `ercot-game-check-csv`. Add `-artifact-dir` when you want the checker to verify scenario bus IDs; `-year`, `-round`, and `-horizon` are optional historical/report context. The validator is a convenience check; the instructor evaluator remains authoritative.

Historical compatibility note. The retired ten-column hourly format with `period` and `load_mw` is retained only for explicitly marked legacy fixtures and compatibility tests. Do not use it for a current submission. The legacy format's asset-ID fields are historical compatibility fields, not part of the active student contract.

5.1 Complete small example

The following is a complete canonical two-site example. It is intentionally small for format testing; a course submission may contain more site rows but still never contains hourly horizon rows:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
team-demo,site-north,1001,4,2,1,4
team-demo,site-south,1002,2,0,0,0
```

The second site has no generation or storage. The half-year hourly scenario is selected by the instructor and is not expanded into repeated rows in this file.

5.2 Checker troubleshooting

- **Unknown bus:** copy the `bus_id` from the supplied ACTIVSg2000 bus catalog.
- **Duplicate site:** give every site row a distinct `site_id`.
- **Storage ratio:** use storage energy equal to four times storage power, or set both storage fields to zero.
- **Malformed CSV:** save as UTF-8 CSV and inspect commas, quotation marks, and the exact header.

6 How joint evaluation works

The instructor first rejects `malformed CSV` files in isolation; a malformed file does not prevent other teams from being evaluated. Review the exact file contract in Section 5 before diagnosing a rejection.

Valid plans are then evaluated jointly. If the joint plan is `feasible`, each team is scored using its cleared hourly load, private resource dispatch, and the announced scoring vectors.

If the joint plan is `infeasible`, the evaluator applies the `fair continuous projection`, discussed in Section 7. No first-come-first-serve ordering is used. The projected load is then used for the score. The capital costs associated with submitted sites, generators, and storage remain part of the score. See Section 8 for the score calculation.

7 Fair continuous projection

Let $d_{i,s,t}$ be the requested load of team i at bus s at time t. When the full joint request is infeasible, the evaluator finds a common continuous service fraction $\alpha \in [0, 1]$ and evaluates the scaled requests

$$\widehat{d}_{i,s,t} = \alpha d_{i,s,t}.$$

The largest feasible α is selected to *maximize the minimum fraction of each team's requested plan that is served.* This policy has three intended properties:

1. every valid team remains in the competition;
2. teams are treated symmetrically at the service-fraction level;
3. the projected dispatch remains a physically feasible DCOPF solution.

The policy is related to `max-min` and proportional fairness ideas used in resource allocation and demand-response research [4, 3, 2]. It is a teaching rule, not an ERCOT operating procedure.

8 Scoring

For team i, the `score` is

$$S_i = V_i^{\mathrm{IT}} - C_i^{\mathrm{energy}} - C_i^{\mathrm{gen,op}} - C_i^{\mathrm{gen,cap}} - C_i^{\mathrm{storage,op}} - C_i^{\mathrm{storage,cap}} - C_i^{\mathrm{site}}.$$

The components are:

- V_i^{IT}: cleared load multiplied by the full-horizon hourly `IT value` vector announced for the round. The current release uses \$125/MWh at every hour;
- C_i^{energy}: grid purchase multiplied by the locational hourly `energy price` artifact for the selected bus. The final artifact is generated from baseline ACTIVSg2000 DCOPF LMPs;
- $C_i^{\mathrm{gen,op}}$: private-generator dispatch multiplied by the dataset-derived marginal cost;

- $C_i^{\text{gen,cap}}$: submitted generator MW multiplied by the announced generator capital coefficient;
- $C_i^{\text{storage,op}}$: announced `storage operating coefficient` multiplied by storage throughput;
- $C_i^{\text{storage,cap}}$: announced storage MW/MWh capital coefficients multiplied by submitted capacities;
- C_i^{site}: number of submitted `connection sites` multiplied by the announced site coefficient.

The current provisional calibration anchors are listed below. The instructor must distribute the final hourly vectors and storage coefficients with the round configuration; do not infer them from this manual if the round file differs.

Parameter	Provisional value	Basis
Generator capital	$85,000/MW	Annualized combustion-turbine anchor from EIA cost data [8].
Site cost	$50,000/site	Instructional interconnection proxy informed by Berkeley Lab interconnection-cost work [7].
Generator operating cost	Piecewise $12/16/24 per MWh	Common team-built curve; same at every site.
Energy price	Baseline LMP artifact	Exogenous ACTIVSg2000 DCOPF price surface, one price per bus and hour.
IT value	$125/MWh hourly	Full-horizon instructor-chosen opportunity value; see the profile documentation.
Storage power capital	$200,000/MW	Rounded power-capacity component calibrated from NREL/EIA battery benchmarks [9, 5].
Storage energy capital	$300,000/MWh	Rounded energy-capacity component; together with power capital gives about $1,400/kW for a four-hour system.
Storage operation	$8/MWh discharged	Variable O&M proxy; the reference fixed O&M value is $40/kW-year [9].
Invalid file score	0	Deterministic grading policy, not a market quantity.

8.1 How to reason about your score

The evaluator first determines how much of your requested load is served. It then adds IT value for served load and subtracts energy, site, generator, and storage costs. A larger plan is therefore not automatically better: an additional site may increase IT value while also increasing energy, congestion, and capital costs. A generator or storage device can reduce grid purchases but still lower the final score if its costs exceed the avoided energy cost.

Use the full-horizon `IT-value` vector and the locational `price` artifact supplied with the round when estimating a score. The current IT-value vector is constant by policy; the energy price is not a student-chosen value.

9 Practical workflow

1. Read the round configuration and identify the evaluator's scenario block.
2. Inspect the ACTIVSg2000 bus and generator tables supplied by the instructor.
3. Choose one or more buses and specify site capacities in the canonical CSV.
4. Decide whether private generation or storage improves your score.
5. Generate the `CSV` and run the `validator`.
6. Keep the exact submitted file and validator output for reproducibility.
7. Submit before the announced deadline.

9.1 What to inspect after evaluation

Read the result in this order:

1. feasibility or projected service status;
2. total score and rank;
3. cleared load, grid purchases, and resource costs;
4. binding or limiting network diagnostics;
5. storage state and generator dispatch;
6. raw audit output only when debugging.

9.2 Final submission checklist

Before uploading, confirm:

- the file is `UTF-8 CSV` with the exact header;
- each data row has a distinct site ID and the exact seven-column header;
- all bus IDs come from the supplied catalog;
- all capacities are finite and nonnegative, with storage energy equal to four times storage power when storage is present;
- the `checker` passes without requiring hourly context arguments;
- you retained the exact submitted file and checker output;
- your strategy note explains the main economic and physical trade-offs.

10 Troubleshooting and frequently asked questions

How are generator and storage assets identified?
The active seven-column CSV has no asset-ID columns, including no generator-ID or storage-ID columns. Submit capacities only; the evaluator assigns deterministic private asset IDs from the team and site IDs. You cannot choose the operating-cost curve: every team-built generator uses the common piecewise curve specified in Section 8, and the evaluator dispatches it.

Can my generator or storage serve another team?
No. These resources are private and can reduce only your team's grid purchase.

What happens if the joint plan is infeasible?
The evaluator first applies the fair continuous projection in Section 7, using one common service fraction for valid teams.

Why is my score negative?
Served-load value may be smaller than energy, site, generator, or storage costs. Inspect the detailed score components before changing the plan.

Which result is official?
The instructor-generated result and audit files determine the official score. Local calculations are for debugging and strategy development.

11 References

References

[1] Caseformat. Activsg2000: Synthetic texas 2000-bus power system model, 2018. Public synthetic network source used by the controlled teaching scenario.

[2] Dalia Gebbran et al. Fairness-aware coordination of distributed energy resources. *IEEE Transactions on Smart Grid*, 2021. Reference for continuous proportional and egalitarian resource allocation.

[3] Doron Jacubowicz, Guy Raz, and Yair Beck. Fairness algorithm for emergency demand response. *International Journal of Electrical Power and Energy Systems*, 134:107871, 2022.

[4] Frank P. Kelly. Charging and rate control for elastic traffic. *European Transactions on Telecommunications*, 8(1):33–37, 1998.

[5] National Renewable Energy Laboratory. Annual technology baseline 2024: Utility-scale battery storage, 2024. Reference for separating battery power and energy capital costs.

[6] Vignesh Ramasamy et al. Annual technology baseline: Utility-scale battery storage. Technical report, National Renewable Energy Laboratory, 2021. Reference for utility-scale four-hour battery capital-cost components.

[7] Joseph Rand et al. Generator interconnection costs to the transmission system. Technical report, Lawrence Berkeley National Laboratory, 2022.

[8] U.S. Energy Information Administration. Capital cost and performance characteristics for utility-scale electric power generating technologies, 2024. Technology-cost reference for the provisional generator capital anchor.

[9] U.S. Energy Information Administration. Capital cost estimates for utility scale electricity generating plants: Battery energy storage system assumptions, 2024. Reference for battery construction, fixed O&M, and variable O&M benchmarks.